

Three lines of PPO reproduce the whole PT-PPO apparatus

pt_full vs vanilla+3 lines: +49 ($p = 0.485$, indistinguishable) · both vs vanilla: $p = 0.002$

50

vanilla PPO

vanilla + 3 lines

pt_full (live permanent)

711

760

551

pt_full (inert permanent)

-250

final-phase return (6 seeds, HalfCheetah, 3.07M steps)

250

500

750

1000

1250

exact permutation test, $p = 0.485$ · seed spread: vanilla 787 vs vanilla+3 lines 49 (16x tighter)